

Class10: Structural Bioinformatics 1

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB).

PDB Statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats obtained from <https://www.rcsb.org/stats/summary>

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					

1	84	32	217,375
2	1	0	14,283
3	0	0	16,277
4	3	1	5,004
5	0	0	244
6	0	4	22

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

80.49% of structures in the PDB are solved by X-Ray and 13.38% by Electron Microscopy.

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

```
pdb$Neutron
```

```
[1] 84 1 0 3 0 0
```

This print out above `pdb$X.ray` is “character” not “numeric”. Therefore I can’t do math with it. We need to fix this...

```
# We want to get rid of (or sub out) commas:
x <- pdb$X.ray
tmp <- sub(",", "", x=pdb$X.ray)
sum( as.numeric(tmp) )
```

```
[1] 203794
```

We could make a function to do this:

```
rm.comma <- function(x) {
  tmp <- gsub(",", "", x)
  sum( as.numeric(tmp) )
}
```

```
n.tot <- rm.comma(pdb$Total)
n.xray <- rm.comma(pdb$X.ray)
n.em <- rm.comma(pdb$EM)

n.xray/n.tot * 100
```

```
[1] 80.48577
```

```
n.em/n.tot * 100
```

```
[1] 13.38046
```

We could also use a different import function `read_csv()` for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file).

```
library(readr)

read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Molecular Type
```

```
dbl (4): Integrative, Multiple methods, Neutron, Other
```

```
num (4): X-ray, EM, NMR, Total
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# A tibble: 6 x 9
```

	`Molecular Type` <chr>	`X-ray` <dbl>	EM <dbl>	NMR <dbl>	Integrative <dbl>	`Multiple methods` <dbl>	Neutron <dbl>
1	Protein (only)	180758	23111	12813	348	229	84
2	Protein/Oligosacch~	10488	3741	34	8	11	1
3	Protein/NA	9205	6751	287	26	8	0
4	Nucleic acid (only)	3154	250	1578	3	15	3
5	Other	178	27	35	4	0	0
6	Oligosaccharide (o~	11	0	6	0	1	0

```
# i 2 more variables: Other <dbl>, Total <dbl>
```

```
n.tot <- sum(as.numeric(gsub(",", "", pdb$Total)))
n.xray <- sum(as.numeric(gsub(",", "", pdb$X.ray)))
n.em <- sum(as.numeric(gsub(",", "", pdb$EM)))

n.xray / n.tot * 100
```

```
[1] 80.48577
```

```
n.em / n.tot * 100
```

```
[1] 13.38046
```

```
(n.xray + n.em) / n.tot * 100
```

```
[1] 93.86623
```

Q. How many total protein structures are there in the dataset?

```
pdb$Total[1]
```

```
[1] "217,375"
```

The total number of protein sequences in UniProt is 202,556,314

```
217315/202556314 * 100
```

```
[1] 0.1072862
```

Key-point: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~%80) come from one method (X-ray crystallography).

Q2: What proportion of structures in the PDB are protein?

```
# Remove commas and convert to numeric
rm.comma <- function(x) {
  tmp <- gsub(",", "", x)
  as.numeric(tmp)
}

# Convert Total column
total <- rm.comma(pdb$Total)

# Protein is the first row
n.protein <- total[1]
n.total <- sum(total)

# Proportion of PDB structures that are protein
n.protein / n.total * 100
```

[1] 85.84941

Approximately 85.9% of structures in the PDB are protein.

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

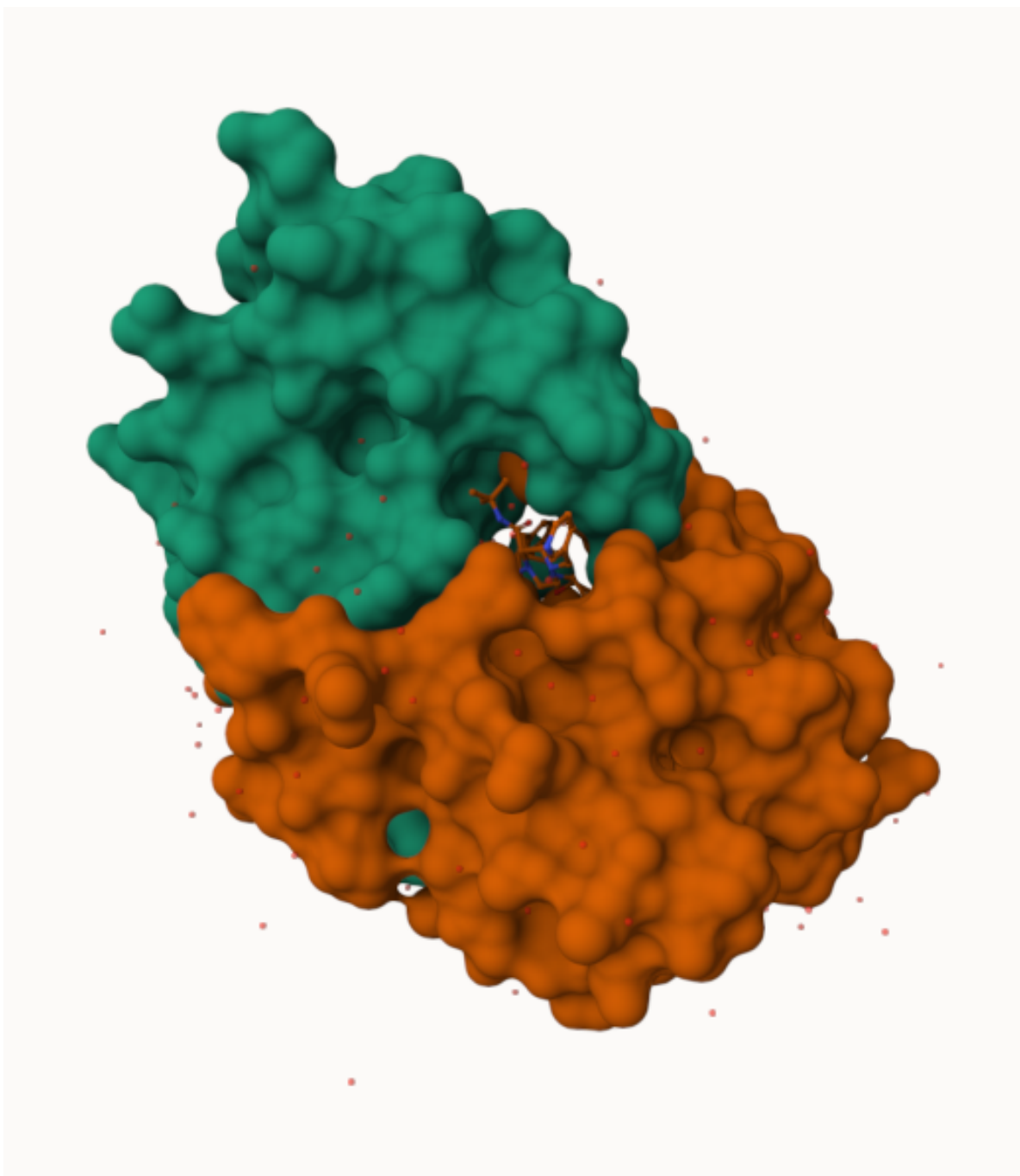
Searching “HIV-1 protease” in the PDB returns 1,227 structures.

Visualizing PDB Data

Main strand alone web version with all features is at <https://molstar.org/viewer/>



Figure 1: The HIV-Protease enzyme is a homodimer with two chains



> Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

In X-ray crystal structures, hydrogen atoms are usually not resolved because they are too small and have low electron density, so only the oxygen atom of each water molecule is visible.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

The critical “conserved” water molecule in the binding site is Residue 308, which bridges interactions within the active site.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.



Figure 2: Spacefill display of catalytic ASP25 amino acids and key binding site water molecule

```
library(bio3d)

# Read the HIV-1 protease structure from the PDB
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD  
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE  
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP  
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

There are 198 residues in this pdb object.

Q8: Name one of the two non-protein residues?

One of the two non-protein residues is water (HOH).

Q9: How many protein chains are in this structure?

There are 2 protein chains in this structure.

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

There are lots of functions that can work with these `pdb` objects:

```
head(pdbseq(pdb))
```

```
  1    2    3    4    5    6  
"P" "Q" "I" "T" "L" "W"
```

```
library(bio3dview)
```

```
view.pdb(pdb)
```

Let's try a custom view

```
view.pdb(pdb,  
  colorScheme="sse",  
  backgroundColor = "black")
```

Q. Create a custom view of HIV-Pr highlighting the active site ASP residues (`resno=25`), the two chains (in your choice of colors), and the ligand all on a custom color background?

```
library(bio3dview)
library(NGLVieweR)

active.site <- atom.select(pdb, resno = 25)

view.pdb(pdb,
  cols = c("red", "blue"),
  highlight = active.site,
  highlight.style = "spacefill",
  backgroundColor = "pink") |>
setRock()
```

Predict the flexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given `pdb` object:

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, `rm.alt=TRUE`

A quick structure summary

```
adk
```

Call: `read.pdb(file = "6s36")`

```
Total Models#: 1
  Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

Protein sequence:

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
TDELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
```

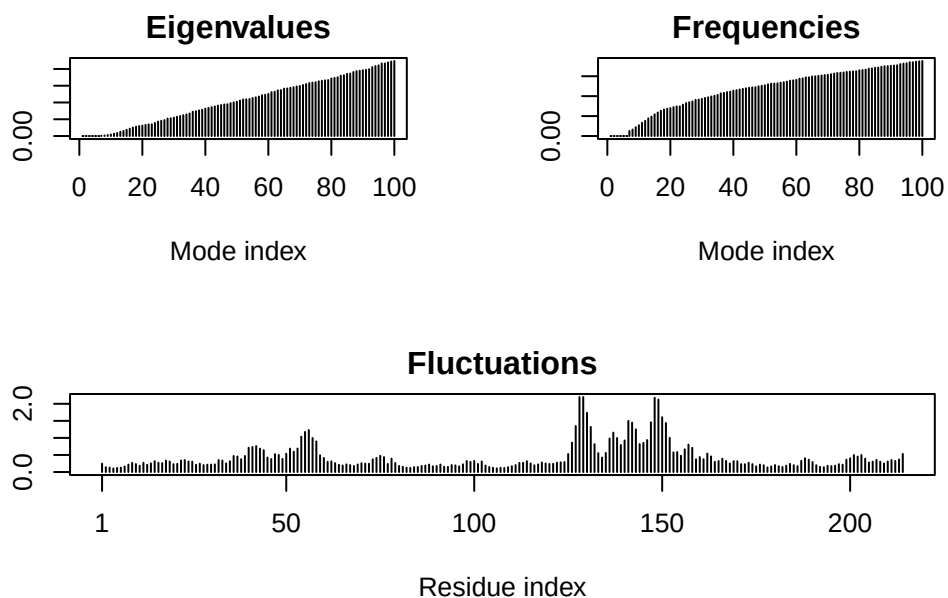
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM TAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
m <- nma(adk)
```

```
Building Hessian...      Done in 0.014 seconds.  
Diagonalizing Hessian... Done in 0.286 seconds.
```

```
plot(m)
```



View the results with an interactive structure view:

```
view.nma(m)
```

Write out the results for viewing in Mol-star:

```
mktrj(m, file = "nma.pdb")
```

You can view quickly here or open the file made previously in Mol-star.

```
view.nma(m, pdb=adk)
```

Comparative analysis of the ADK family

```
# Install packages in the R console NOT your Rmd/Quarto file
# install.packages("bio3d")
# install.packages("NGLViewer")
```

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa is found on BioConductor and not CRAN.

Q11. Which of the above packages is not found on BioConductor or CRAN?:

bio3dview is not found on CRAN or BioConductor because it is installed directly from GitHub using `remotes::install_github()`.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True.

Our first step is find a sequence for this family. We will use the database iD “1ake_A” here:

```
id <- "1ake_A"
aa <- get.seq(id)
```

Warning in `get.seq(id)`: Removing existing file: `seqs.fasta`

Fetching... Please wait. Done.

```
aa
```

	1	.	.	.	.	.	60
pdb 1AKE A	MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV						
	1	.	.	.	.	.	60
	61	.	.	.	.	.	120
pdb 1AKE A	DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI						
	61	.	.	.	.	.	120

```

      121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      121      .      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214

```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

There are 214 amino acids in this sequence.

Search for related sequences in the database

```
blast <- blast.pdb(aa)
```

```
Searching ... please wait (updates every 5 seconds) RID = ZRG2YM6J016
```

```
.....
```

```
Reporting 96 hits
```

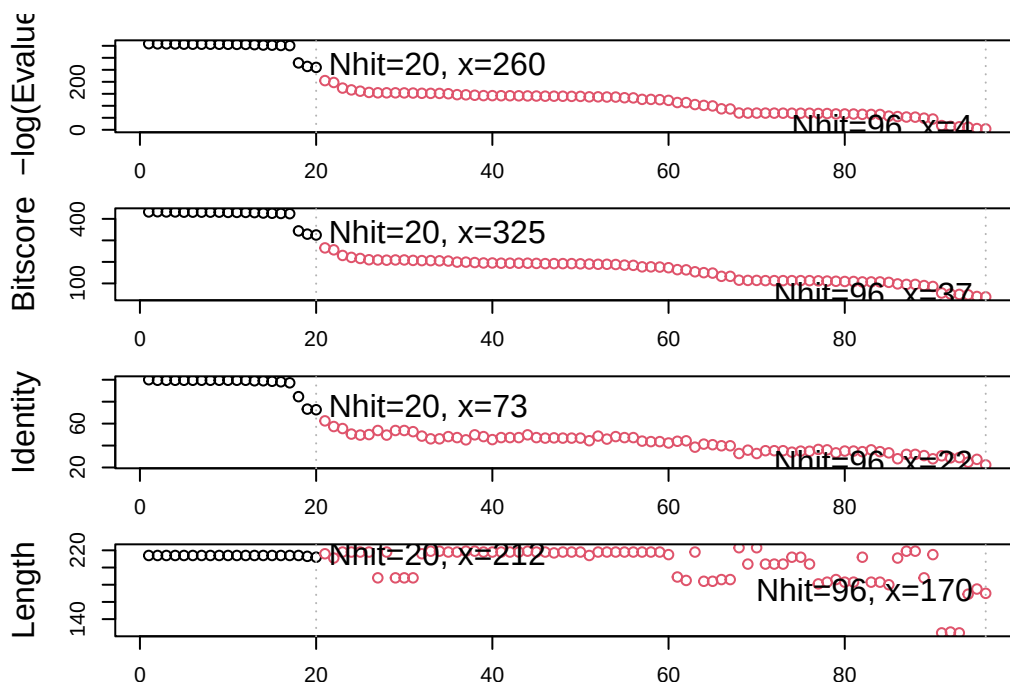
```
hits <- plot(blast)
```

```
* Possible cutoff values: 260 3
```

```
Yielding Nhits: 20 96
```

```
* Chosen cutoff value of: 260
```

```
Yielding Nhits: 20
```



```
hits$ pdb.id
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
[9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
# Download related PDB files
files <- get.pdb(hits$ pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6S36.pdb.gz exists. Skipping download
```


Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

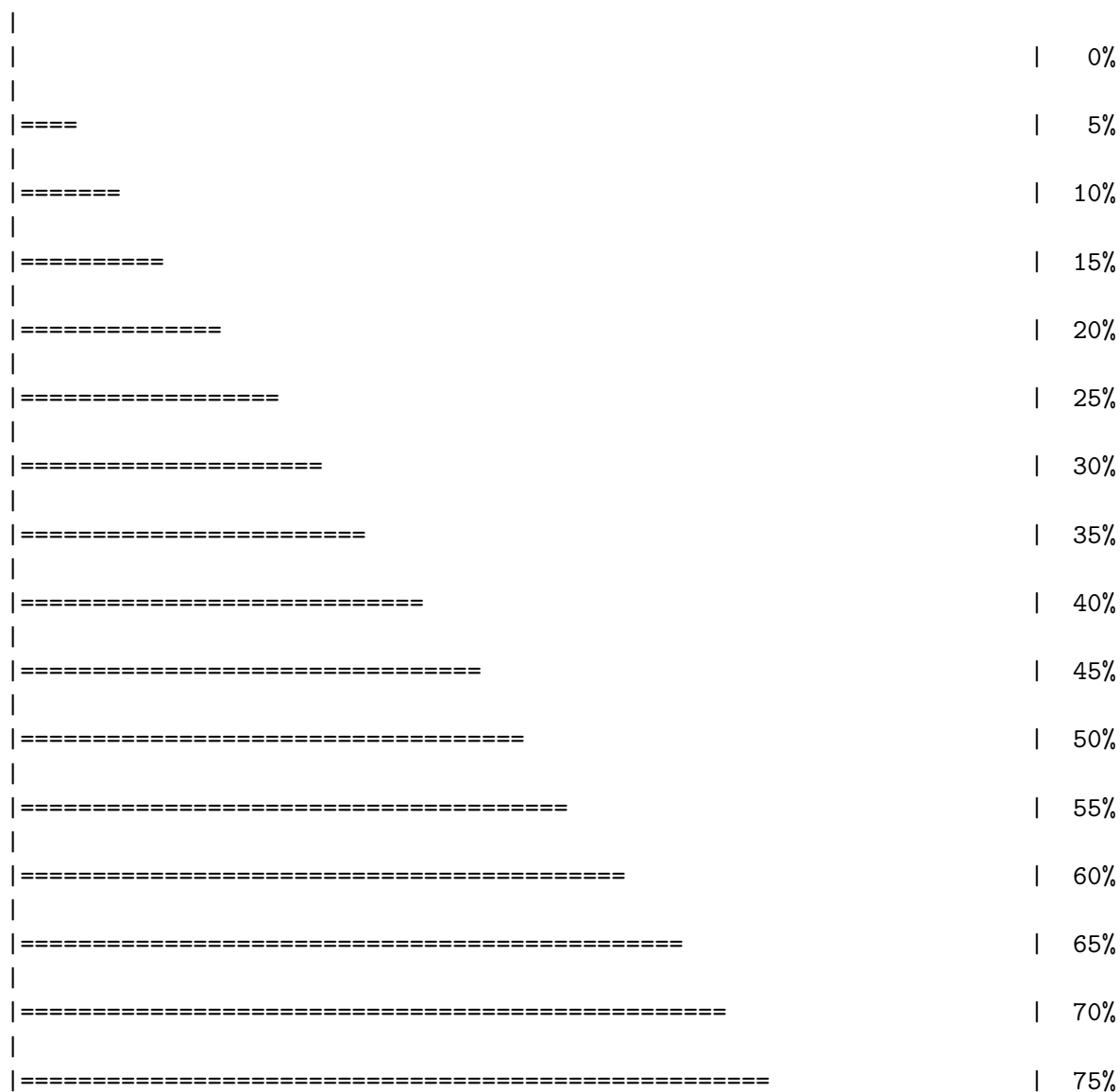
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

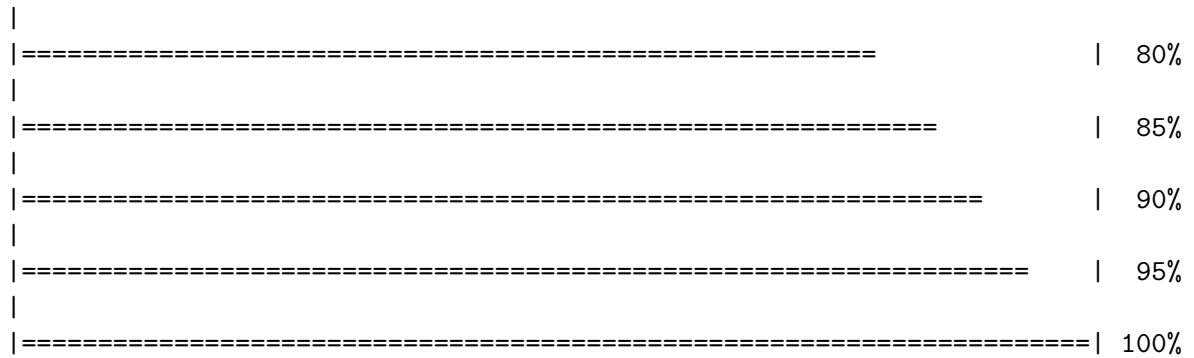
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download





Align and superpose all these ADK structures

```
library(msa)
```

Loading required package: Biostrings

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

```
IQR, mad, sd, var, xtabs
```

The following objects are masked from 'package:base':

```
anyDuplicated, aperm, append, as.data.frame, basename, cbind,  
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,  
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,  
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,  
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,  
unsplit, which.max, which.min
```

Loading required package: S4Vectors

Loading required package: stats4

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

```
findMatches
```

The following objects are masked from 'package:base':

```
expand.grid, I, unname
```

Loading required package: IRanges

Attaching package: 'IRanges'

The following object is masked from 'package:bio3d':

```
trim
```

Loading required package: XVector

Loading required package: Seqinfo

Attaching package: 'Biostrings'

The following object is masked from 'package:bio3d':

mask

The following object is masked from 'package:base':

strsplit

```
# Align related PDBs
pdds <- pddaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdds/split_chain/1AKE_A.pdb
pdds/split_chain/8BQF_A.pdb
pdds/split_chain/4X8M_A.pdb
pdds/split_chain/6S36_A.pdb
pdds/split_chain/9R6U_A.pdb
pdds/split_chain/9R71_A.pdb
pdds/split_chain/8Q2B_A.pdb
pdds/split_chain/8RJ9_A.pdb
pdds/split_chain/6RZE_A.pdb
pdds/split_chain/4X8H_A.pdb
pdds/split_chain/3HPR_A.pdb
pdds/split_chain/1E4V_A.pdb
pdds/split_chain/5EJE_A.pdb
pdds/split_chain/1E4Y_A.pdb
pdds/split_chain/3X2S_A.pdb
pdds/split_chain/6HAP_A.pdb
pdds/split_chain/6HAM_A.pdb
pdds/split_chain/8PVW_A.pdb
pdds/split_chain/4K46_A.pdb
pdds/split_chain/4NP6_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
```

```
.... PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
..
```

Extracting sequences

```
pdb/seq: 1 name: pdbc/split_chain/1AKE_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2 name: pdbc/split_chain/8BQF_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3 name: pdbc/split_chain/4X8M_A.pdb
pdb/seq: 4 name: pdbc/split_chain/6S36_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5 name: pdbc/split_chain/9R6U_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6 name: pdbc/split_chain/9R71_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7 name: pdbc/split_chain/8Q2B_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8 name: pdbc/split_chain/8RJ9_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9 name: pdbc/split_chain/6RZE_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10 name: pdbc/split_chain/4X8H_A.pdb
pdb/seq: 11 name: pdbc/split_chain/3HPR_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12 name: pdbc/split_chain/1E4V_A.pdb
pdb/seq: 13 name: pdbc/split_chain/5EJE_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14 name: pdbc/split_chain/1E4Y_A.pdb
pdb/seq: 15 name: pdbc/split_chain/3X2S_A.pdb
pdb/seq: 16 name: pdbc/split_chain/6HAP_A.pdb
pdb/seq: 17 name: pdbc/split_chain/6HAM_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18 name: pdbc/split_chain/8PVW_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19 name: pdbc/split_chain/4K46_A.pdb
PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 20 name: pdbc/split_chain/4NP6_A.pdb
```

[Truncated_Name:1] 1AKE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:2] 8BQF_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:3] 4X8M_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:4] 6S36_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:5] 9R6U_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:6] 9R71_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:7] 8Q2B_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:8] 8RJ9_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:9] 6RZE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:10] 4X8H_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:11] 3HPR_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:12] 1E4V_A.pdb	--MRIILLGAPVAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:13] 5EJE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:14] 1E4Y_A.pdb	--MRIILLGALVAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:15] 3X2S_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:16] 6HAP_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:17] 6HAM_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:18] 8PVW_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:19] 4K46_A.pdb	--MRIILLGAPGAGKGTQAQFIMAKFGIPQISTGDMLRAA
[Truncated_Name:20] 4NP6_A.pdb	NAMRIILLGAPGAGKGTQAQFIMEKFIPQISTGDMLRAA
	***** ***** *~*****
1	.
41	.
[Truncated_Name:1] 1AKE_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:2] 8BQF_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQE---
[Truncated_Name:3] 4X8M_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:4] 6S36_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:5] 9R6U_A.pdb	VKSGSELGAQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:6] 9R71_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:7] 8Q2B_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:8] 8RJ9_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:9] 6RZE_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:10] 4X8H_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:11] 3HPR_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:12] 1E4V_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:13] 5EJE_A.pdb	VKSGSELGKQAKDIMDACKLVTDELVIALVKERIAQEDCR
[Truncated_Name:14] 1E4Y_A.pdb	VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:15] 3X2S_A.pdb	VKSGSELGKQAKDIMDCGKLVTDELVIALVKERIAQEDSR

[Truncated_Name:16] 6HAP_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVRERICQEDSR		
[Truncated_Name:17] 6HAM_A.pdb	IKSGSELGKQAKDIMDAGKLVDEIIIALVKERICQEDSR		
[Truncated_Name:18] 8PVW_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR		
[Truncated_Name:19] 4K46_A.pdb	IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA		
[Truncated_Name:20] 4NP6_A.pdb	IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE		
	^* *~*** *** ^~* **~*~^~^*~^*** *		
	41 . . .		80
	81 . . .		120
[Truncated_Name:1] 1AKE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:2] 8BQF_A.pdb	-GFLLDGFPR TIPQADAMKEAGINVDYVIEFDVPDELIVD		
[Truncated_Name:3] 4X8M_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:4] 6S36_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:5] 9R6U_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:6] 9R71_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDALIVD		
[Truncated_Name:7] 8Q2B_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:8] 8RJ9_A.pdb	NGFLLAGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:9] 6RZE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:10] 4X8H_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:11] 3HPR_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:12] 1E4V_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:13] 5EJE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:14] 1E4Y_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:15] 3X2S_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:16] 6HAP_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:17] 6HAM_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:18] 8PVW_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:19] 4K46_A.pdb	KGFLLDGFPR TIPQADGLKEVGVVVDYVIEFDVADSVIVE		
[Truncated_Name:20] 4NP6_A.pdb	KGFLLDGFPR TIPQADGLKEMGINVDYVIEFDVADDVIVE		
	**** *****^~** *^ *****^***** * ^**^		
	81 . . .		120
	121 . . .		160
[Truncated_Name:1] 1AKE_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:2] 8BQF_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:3] 4X8M_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:4] 6S36_A.pdb	KIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:5] 9R6U_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:6] 9R71_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:7] 8Q2B_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKA		
[Truncated_Name:8] 8RJ9_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:9] 6RZE_A.pdb	AIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:10] 4X8H_A.pdb	RIVGRRVHAPSGRVYHV KFNPPKVEGKDDVTGEELTTRKD		

[Truncated_Name:11] 3HPR_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDGTGEELTTRKD	
[Truncated_Name:12] 1E4V_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:13] 5EJE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:14] 1E4Y_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:15] 3X2S_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:16] 6HAP_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:17] 6HAM_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:18] 8PVW_A.pdb	RILKR--GETSGRV-----D	
[Truncated_Name:19] 4K46_A.pdb	RMAGRRAHLASGRTYHNVNPPKVEGKDDVTGEDLVIRE	
[Truncated_Name:20] 4NP6_A.pdb	RMAGRRAHLPSGRTYHVYNPPKVEGKDDVTGEDLVIRE	
	^ * ***	
	121 . . .	160
	161 . . .	200
[Truncated_Name:1] 1AKE_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:2] 8BQF_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:3] 4X8M_A.pdb	DQEETVRKRLVEWHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:4] 6S36_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:5] 9R6U_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:6] 9R71_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:7] 8Q2B_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:8] 8RJ9_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:9] 6RZE_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:10] 4X8H_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:11] 3HPR_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:12] 1E4V_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:13] 5EJE_A.pdb	DQEECVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:14] 1E4Y_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:15] 3X2S_A.pdb	DQEETVRKRLCEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:16] 6HAP_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:17] 6HAM_A.pdb	DQEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:18] 8PVW_A.pdb	DNEETVRKRLVEYHQM TAPLIGYYSKEAEAGNTKYAKVDG	
[Truncated_Name:19] 4K46_A.pdb	DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG	
[Truncated_Name:20] 4NP6_A.pdb	DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG	
	* * * * * ^ * * * * * * * * * *	
	161 . . .	200
	201 . 216	
[Truncated_Name:1] 1AKE_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:2] 8BQF_A.pdb	TKPVAEVRADLEKIL-	
[Truncated_Name:3] 4X8M_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:4] 6S36_A.pdb	TKPVAEVRADLEKILG	
[Truncated_Name:5] 9R6U_A.pdb	TKPVAEVRADLEKILG	

```

[Truncated_Name:6] 9R71_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:7] 8Q2B_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:8] 8RJ9_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:9] 6RZE_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:10] 4X8H_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:11] 3HPR_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:12] 1E4V_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:13] 5EJE_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:14] 1E4Y_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:15] 3X2S_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:16] 6HAP_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:17] 6HAM_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:18] 8PVW_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:19] 4K46_A.pdb     TKAVAEVSAELEKALA
[Truncated_Name:20] 4NP6_A.pdb     TKQVSEVSADIAKALA
                                ** * ** *~* *
                                201      .      216

```

Call:

```
pdbaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

```
pdbs, fasta
```

Alignment dimensions:

```
20 sequence rows; 216 position columns (182 non-gap, 34 gap)
```

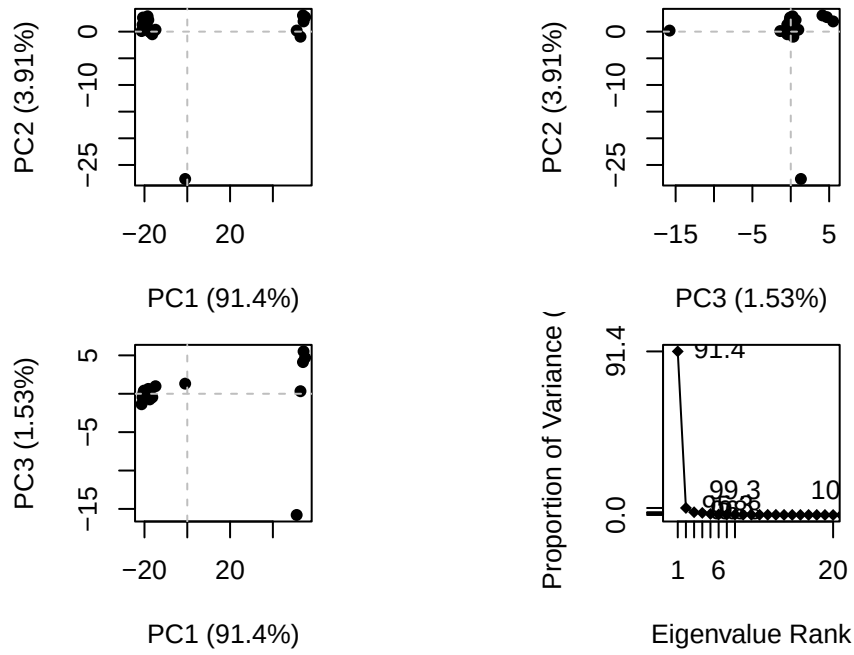
```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

Quick interactive structural view

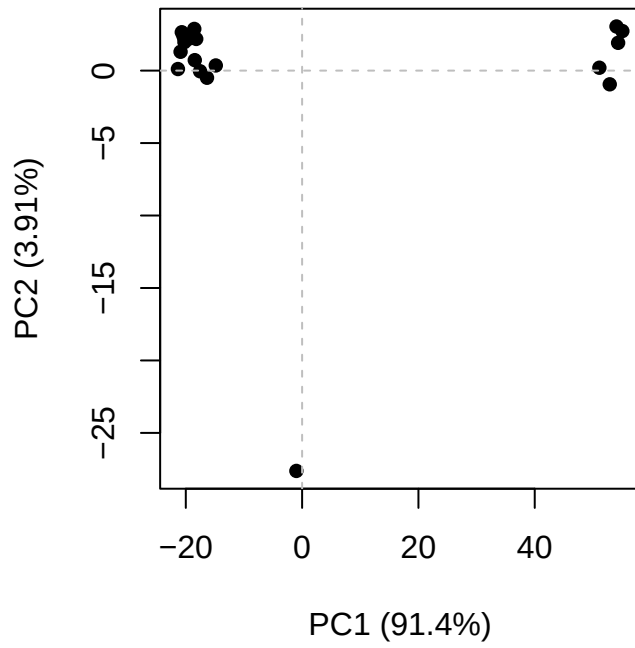
```
view.pdbs(pdbs)
```

PCA of all this structural data (x, y, and z atom coordinates):

```
pc <- pca(pdbs)
plot(pc)
```



```
plot(pc, 1:2)
```



Interactive view of the PC1 captured structural differences:

```
view.pca(pc)
```

```
mktrj(pc, file = "pca.pdb")
```